VECTOR BUNDLES FOR THE COLIMIT ALONG FROBENIUS FORM A STACK IN GROUPOIDS FOR THE H TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EXE (sources/stacks-flat/pinned-original-source.tex line 12440).

Allowed standard prerequisites (editorial).

- The stack conditions to be checked. Same chapter, Lemma labelled lemma-refine-check-h-stack, cited at proof lines 12456-12458. Lists conditions (1), (2) and (3).
- fppf descent of finite locally free modules. Descent, Proposition labelled descent-proposition-fpqc-descent-quasi-coherent and Lemma labelled descent-lemma-finite-locally-free-descends, cited at proof lines 12468-12472. Supplies the equivalence of categories verifying condition (2).
- The fibre category as a colimit of vector bundle categories. Same chapter, Lemma labelled lemma-colim-F-Vect, cited at proof lines 12452-12454. Gives the description of the fibre categories used throughout.
- The two almost blowup computations. Same chapter, Lemmas labelled lemma-vector-bundle-I and lemma-vector-bundle-II, cited at proof lines 12478-12483. Settle condition (3).

Source reference. Original lemma 0EXB: sources/stacks-flat/pinned-original-source.tex, source lines 12223–12223. Same-chapter pointer into the author environment lemma at source line 12223; invoked at proof line 12453 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EX9: sources/stacks-flat/pinned-original-source.tex, source lines 11696–11696. Same-chapter pointer into the author environment lemma at source line 11696; invoked at proof line 12455 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 023T: sources/stacks-chow/tags.txt, source lines 2754–2754. Cross-chapter standard prerequisite located in the project tag list

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staged in the local snapshot; invoked at proof line 12477 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 05B2: sources/stacks-chow/tags.txt, source lines 6870–6870. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12478 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EXC: sources/stacks-flat/pinned-original-source.tex, source lines 12235–12235. Same-chapter pointer into the author environment lemma at source line 12235; invoked at proof line 12482 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EXD: sources/stacks-flat/pinned-original-source.tex, source lines 12303–12303. Same-chapter pointer into the author environment lemma at source line 12303; invoked at proof line 12482 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Proposition 0.1. *Let p be a prime number. Let S be a scheme in characteristic p. Then the category fibred in groupoids*

$$p : \mathcal{S} \longrightarrow (Sch/S)_h$$

whose fibre category over U is the category of finite locally free $\mathrm{colim}_F \mathcal{O}_U$*-modules over U is a stack in groupoids. Moreover, if U is quasi-compact and quasi-separated, then $\mathcal{S}_U$ is* colim_F *Vect*(U).

Proof. The final assertion is the content of Lemma 0EXB. To prove the proposition we will check conditions (1), (2), and (3) of Lemma 0EX9.

Condition (1) holds because by definition we have glueing for the Zariski topology.

To see condition (2), suppose that $f : X \to Y$ is a surjective, flat, proper morphism of finite presentation over S with Y affine. Since $Y, X, X \times_Y X$ are quasi-compact and quasi-separated, we can use the description of fibre categories given in the statement of the proposition. Then it is clearly enough to show that

$$\mathit{Vect}(Y) \longrightarrow \mathit{Vect}(X) \times_{\mathit{Vect}(X \times_Y X)} \mathit{Vect}(X)$$

is an equivalence (as this will imply the same for the colimits). This follows immediately from fppf descent of finite locally free modules, see Descent, Proposition 023T and Lemma 05B2.

Condition (3) is the content of Lemmas 0EXC and 0EXD. □